

# lasso\_psm\_analysis

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## 1. データ読み込み

```
# =====  
# 0. 初期設定・パッケージ読み込み  
# =====  
library(tinytex)  
library(here)
```

here() starts at /Users/snakada/Documents/univ\_github/peru\_endes

```
library(haven)  
library(glmnet)
```

Loading required package: Matrix

Loaded glmnet 4.1-10

```
library(tidyverse)
```

```
-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --  
v dplyr   1.1.4   v readr   2.1.5  
v forcats 1.0.0   v stringr 1.5.1  
v ggplot2 3.5.2   v tibble  3.3.0  
v lubridate 1.9.4 v tidyr   1.3.1  
v purrr    1.0.4
```

```
-- Conflicts ----- tidyverse_conflicts() --
x tidyr::expand() masks Matrix::expand()
x dplyr::filter() masks stats::filter()
x dplyr::lag() masks stats::lag()
x tidyr::pack() masks Matrix::pack()
x tidyr::unpack() masks Matrix::unpack()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# library(fastDummies)
library(nnet) # 多項ロジスティック回帰
library(survey) # 重み付き統計解析
```

Loading required package: grid  
Loading required package: survival

Attaching package: 'survey'

The following object is masked from 'package:graphics':

dotchart

```
library(did) # Staggered DID用（必要に応じて）
library(sjlabelled) # ラベル付きデータ処理用
```

Attaching package: 'sjlabelled'

The following object is masked from 'package:forcats':

as\_factor

The following object is masked from 'package:dplyr':

as\_label

The following object is masked from 'package:ggplot2':

as\_label

The following objects are masked from 'package:haven':

as\_factor, read\_sas, read\_spss, read\_stata, write\_sas, zap\_labels

```

library(tableone)
library(fixest) # 回帰分析の推定結果から係数表（推定値、標準誤差、t値、p値など）を抽出

# オブジェクトクリア
rm(list = ls(all = TRUE))

# デバッグモードフラグ
debug_mode = TRUE # デバッグモード
# debug_mode = FALSE # 通常モード

# 自作関数読み込み
source("../myTools.R")

```

Attaching package: 'psych'

The following object is masked from 'package:did':

sim

The following objects are masked from 'package:ggplot2':

%+%, alpha

```

# =====
# 1. データ読み込み
# =====
# DHSデータのルートフォルダを指定
gdrive_dir <- "/Users/snakada/Library/CloudStorage/GoogleDrive-snakada@g.ecc.u-tokyo.ac.jp/マイドライブ/Peru_work/"

# 結果保存先
save_path <- file.path(gdrive_dir, "output", "lasso_stunting")

df_org <- readRDS(
  file.path(gdrive_dir, "output", "all_data_merged.rds")
) %>% filter(!is.na(nt_ch_stunt))

# データベース一覧
endes_list <- readRDS(file.path(gdrive_dir, "output", "endes_data_list.rds"))

# 変数ラベル一覧

```

```

endes_var_labels <- readRDS(file.path(gdrive_dir, "output", "endes_var_labels.rds"))

# 介入群グループ化
df_org <- df_org %>%
  mutate(
    treatment_group = case_when(
      treatment_start == "2009" ~ 1, # 早期介入
      treatment_start == "2011" ~ 2, # 後期介入
      TRUE ~ 0 # コントロール
    ),
    treatment_group = as.factor(treatment_group)
  )
# 変数ラベル取得
var_labels <- df_org %>% get_label %>% as_tibble() %>% rename(labels = value)
var_labels$varname <- names(df_org)

```

## 2. LASSO用データ前処理（介入前データのみ）

```

# =====
# 2. LASSO用データ前処理（介入前データのみ）
# =====

library(gt)

# 除外変数定義(初期値)
drop_cols_init <- c(
  "b3", "b4", "b5", "b9", "bidx", "caseid", "hc27", "hhid", "m1", "m13", "m14", "m15", "m1d",
  "m2a", "m2b", "m2c", "m2d", "m2e", "m2f", "m2g", "m2h", "m2i", "m2j", "m2k", "m2l", "m2m",
  "m3a", "m3b", "m3c", "m3d", "m3e", "m3f", "m3g", "m3h", "m3i", "m3j", "m3k", "m3l", "m3m",
  "m3n", "m42c", "m42d", "m42e", "m43", "m45", "m60", "m61", "v001", "v002", "v003", "v005",
  "v008", "v012", "v013", "v021", "v022", "v024", "v025", "v026", "v106", "v190", "v208",
  "v467b", "v467c", "v467d", "v467f",
  "wt", "state", "age", "treated_status", "treatment_start", "year_treated",
  "ch_report_bw", "ch_report_bw", "ch_size_birth", "dm_cooking_fuel_traditional", "dm_cooking_fuel_traditional", "dm_rural", "dm_rural",
  "rh_anc_bldpres", "rh_anc_median", "rh_anc_bldsamp", "rh_anc_median", "rh_anc_numvs", "rh_anc_numvsrh_anc_moprg", "rh_anc_moprg",
  )

# 目的変数と同値となる除去変数("nt_ch_sev_stunt", "nt_ch_haz", "nt_ch_stunt"のうち2つ)
y_var_all <- c("nt_ch_stunt", "nt_ch_sev_stunt", "nt_ch_haz", "nt_ch_sev_wast", "nt_ch_wast", "nt_ch_whz", "nt_ch_owwt_ht", "nt_ch_owwt_ht")
y_var <- "nt_ch_stunt"

```

```

y_var_drop <- setdiff(y_var_all, y_var)

# 除外変数定義(最終値)
drop_cols <- c(drop_cols_init, y_var_all)

# ID・管理用変数
grouping_cols <- c("caseid", "state", "treated_status", "treatment_start", "group_treated",
                  "wt", "treatment_group", y_var)

# 1. LASSO用のデータセット定義
df_lasso <- df_org

cat("介入前データの群別サンプル数:\n")

```

介入前データの群別サンプル数:

```
print(table(df_lasso$treatment_group))
```

```

  0    1    2
24737 3543 3792

```

```

# 4. NAの多い列を除去（10%以上NA）
initial_cols <- ncol(df_lasso)
keep_cols <- names(df_lasso)[sapply(df_lasso, function(x) sum(is.na(x)) <= nrow(df_lasso) * 0.1)]
df_lasso <- df_lasso[, keep_cols]
cat("NA値の多い列を", initial_cols - ncol(df_lasso), "個削除しました\n")

```

NA値の多い列を 139 個削除しました

```

# 5. 共変量名の再定義（NA除去後）
all_drop_cols <- intersect(drop_cols, names(df_lasso))
grouping_cols <- intersect(grouping_cols, names(df_lasso))
covariate_names <- setdiff(names(df_lasso), c(all_drop_cols, grouping_cols))

# 6. 共変量のNAがある行を除去
# df_lasso <- df_lasso[, covariate_names]
complete_idx <- complete.cases(df_lasso)
initial_rows <- nrow(df_lasso)

```

```
df_lasso <- df_lasso[complete_idx, ]
cat("NA値を含む行を", initial_rows - nrow(df_lasso), "行削除しました\n")
```

NA値を含む行を 7958 行削除しました

```
cat("最終的な介入前データサイズ:", nrow(df_lasso), "行 ×", ncol(df_lasso), "列\n\n")
```

最終的な介入前データサイズ: 24114 行 × 182 列

```
# 共変量のラベル保存
covariate_names_label <- var_labels %>%
  filter(varname %in% covariate_names)
# print文の代替
covariate_names_label %>%
  gt() %>%
  cols_width(
    1 ~ pct(80),
    2 ~ pct(20)
  ) %>%
  print()
```

```
<div id="hitsdzdqpj" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;overflow-x:auto;overflow-y:auto;width:auto;height:auto;">
  <style>#hitsdzdqpj table {
    font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Segoe UI Emoji', 'Segoe UI Symbol';
    -webkit-font-smoothing: antialiased;
    -moz-osx-font-smoothing: grayscale;
  }

  #hitsdzdqpj thead, #hitsdzdqpj tbody, #hitsdzdqpj tfoot, #hitsdzdqpj tr, #hitsdzdqpj td, #hitsdzdqpj th {
    border-style: none;
  }

  #hitsdzdqpj p {
    margin: 0;
    padding: 0;
  }

  #hitsdzdqpj .gt_table {
    display: table;
  }
```

```
border-collapse: collapse;
line-height: normal;
margin-left: auto;
margin-right: auto;
color: #333333;
font-size: 16px;
font-weight: normal;
font-style: normal;
background-color: #FFFFFF;
width: auto;
border-top-style: solid;
border-top-width: 2px;
border-top-color: #A8A8A8;
border-right-style: none;
border-right-width: 2px;
border-right-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #A8A8A8;
border-left-style: none;
border-left-width: 2px;
border-left-color: #D3D3D3;
}
```

```
#hitsdzdqpj .gt_caption {
padding-top: 4px;
padding-bottom: 4px;
}
```

```
#hitsdzdqpj .gt_title {
color: #333333;
font-size: 125%;
font-weight: initial;
padding-top: 4px;
padding-bottom: 4px;
padding-left: 5px;
padding-right: 5px;
border-bottom-color: #FFFFFF;
border-bottom-width: 0;
}
```

```
#hitsdzdqpj .gt_subtitle {
color: #333333;
```

```
font-size: 85%;  
font-weight: initial;  
padding-top: 3px;  
padding-bottom: 5px;  
padding-left: 5px;  
padding-right: 5px;  
border-top-color: #FFFFFF;  
border-top-width: 0;  
}
```

```
#hitsdzdqpj .gt_heading {  
  background-color: #FFFFFF;  
  text-align: center;  
  border-bottom-color: #FFFFFF;  
  border-left-style: none;  
  border-left-width: 1px;  
  border-left-color: #D3D3D3;  
  border-right-style: none;  
  border-right-width: 1px;  
  border-right-color: #D3D3D3;  
}
```

```
#hitsdzdqpj .gt_bottom_border {  
  border-bottom-style: solid;  
  border-bottom-width: 2px;  
  border-bottom-color: #D3D3D3;  
}
```

```
#hitsdzdqpj .gt_col_headings {  
  border-top-style: solid;  
  border-top-width: 2px;  
  border-top-color: #D3D3D3;  
  border-bottom-style: solid;  
  border-bottom-width: 2px;  
  border-bottom-color: #D3D3D3;  
  border-left-style: none;  
  border-left-width: 1px;  
  border-left-color: #D3D3D3;  
  border-right-style: none;  
  border-right-width: 1px;  
  border-right-color: #D3D3D3;  
}
```



```
#hitsdzdqpj .gt_col_heading {
  color: #333333;
  background-color: #FFFFFF;
  font-size: 100%;
  font-weight: normal;
  text-transform: inherit;
  border-left-style: none;
  border-left-width: 1px;
  border-left-color: #D3D3D3;
  border-right-style: none;
  border-right-width: 1px;
  border-right-color: #D3D3D3;
  vertical-align: bottom;
  padding-top: 5px;
  padding-bottom: 6px;
  padding-left: 5px;
  padding-right: 5px;
  overflow-x: hidden;
}
```

```
#hitsdzdqpj .gt_column_spanner_outer {
  color: #333333;
  background-color: #FFFFFF;
  font-size: 100%;
  font-weight: normal;
  text-transform: inherit;
  padding-top: 0;
  padding-bottom: 0;
  padding-left: 4px;
  padding-right: 4px;
}
```

```
#hitsdzdqpj .gt_column_spanner_outer:first-child {
  padding-left: 0;
}
```

```
#hitsdzdqpj .gt_column_spanner_outer:last-child {
  padding-right: 0;
}
```

```
#hitsdzdqpj .gt_column_spanner {
  border-bottom-style: solid;
  border-bottom-width: 2px;
}
```

```
border-bottom-color: #D3D3D3;
vertical-align: bottom;
padding-top: 5px;
padding-bottom: 5px;
overflow-x: hidden;
display: inline-block;
width: 100%;
}
```

```
#hitsdzdqj .gt_spanner_row {
border-bottom-style: hidden;
}
```

```
#hitsdzdqj .gt_group_heading {
padding-top: 8px;
padding-bottom: 8px;
padding-left: 5px;
padding-right: 5px;
color: #333333;
background-color: #FFFFFF;
font-size: 100%;
font-weight: initial;
text-transform: inherit;
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
border-left-style: none;
border-left-width: 1px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 1px;
border-right-color: #D3D3D3;
vertical-align: middle;
text-align: left;
}
```

```
#hitsdzdqj .gt_empty_group_heading {
padding: 0.5px;
color: #333333;
background-color: #FFFFFF;
```

```

font-size: 100%;
font-weight: initial;
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
vertical-align: middle;
}

#hitsdzdqpj.gt_from_md > :first-child {
  margin-top: 0;
}

#hitsdzdqpj.gt_from_md > :last-child {
  margin-bottom: 0;
}

#hitsdzdqpj.gt_row {
  padding-top: 8px;
  padding-bottom: 8px;
  padding-left: 5px;
  padding-right: 5px;
  margin: 10px;
  border-top-style: solid;
  border-top-width: 1px;
  border-top-color: #D3D3D3;
  border-left-style: none;
  border-left-width: 1px;
  border-left-color: #D3D3D3;
  border-right-style: none;
  border-right-width: 1px;
  border-right-color: #D3D3D3;
  vertical-align: middle;
  overflow-x: hidden;
}

#hitsdzdqpj.gt_stub {
  color: #333333;
  background-color: #FFFFFF;
  font-size: 100%;
  font-weight: initial;

```

```
text-transform: inherit;
border-right-style: solid;
border-right-width: 2px;
border-right-color: #D3D3D3;
padding-left: 5px;
padding-right: 5px;
}
```

```
#hitsdzdqpj .gt_stub_row_group {
  color: #333333;
  background-color: #FFFFFF;
  font-size: 100%;
  font-weight: initial;
  text-transform: inherit;
  border-right-style: solid;
  border-right-width: 2px;
  border-right-color: #D3D3D3;
  padding-left: 5px;
  padding-right: 5px;
  vertical-align: top;
}
```

```
#hitsdzdqpj .gt_row_group_first td {
  border-top-width: 2px;
}
```

```
#hitsdzdqpj .gt_row_group_first th {
  border-top-width: 2px;
}
```

```
#hitsdzdqpj .gt_summary_row {
  color: #333333;
  background-color: #FFFFFF;
  text-transform: inherit;
  padding-top: 8px;
  padding-bottom: 8px;
  padding-left: 5px;
  padding-right: 5px;
}
```

```
#hitsdzdqpj .gt_first_summary_row {
  border-top-style: solid;
  border-top-color: #D3D3D3;
}
```

```

}

#hitsdzdqj .gt_first_summary_row.thick {
    border-top-width: 2px;
}

#hitsdzdqj .gt_last_summary_row {
    padding-top: 8px;
    padding-bottom: 8px;
    padding-left: 5px;
    padding-right: 5px;
    border-bottom-style: solid;
    border-bottom-width: 2px;
    border-bottom-color: #D3D3D3;
}

#hitsdzdqj .gt_grand_summary_row {
    color: #333333;
    background-color: #FFFFFF;
    text-transform: inherit;
    padding-top: 8px;
    padding-bottom: 8px;
    padding-left: 5px;
    padding-right: 5px;
}

#hitsdzdqj .gt_first_grand_summary_row {
    padding-top: 8px;
    padding-bottom: 8px;
    padding-left: 5px;
    padding-right: 5px;
    border-top-style: double;
    border-top-width: 6px;
    border-top-color: #D3D3D3;
}

#hitsdzdqj .gt_last_grand_summary_row_top {
    padding-top: 8px;
    padding-bottom: 8px;
    padding-left: 5px;
    padding-right: 5px;
    border-bottom-style: double;
    border-bottom-width: 6px;
}

```

```

border-bottom-color: #D3D3D3;
}

#hitsdzdqpj .gt_stripped {
background-color: rgba(128, 128, 128, 0.05);
}

#hitsdzdqpj .gt_table_body {
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
}

#hitsdzdqpj .gt_footnotes {
color: #333333;
background-color: #FFFFFF;
border-bottom-style: none;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
border-left-style: none;
border-left-width: 2px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 2px;
border-right-color: #D3D3D3;
}

#hitsdzdqpj .gt_footnote {
margin: 0px;
font-size: 90%;
padding-top: 4px;
padding-bottom: 4px;
padding-left: 5px;
padding-right: 5px;
}

#hitsdzdqpj .gt_sourcenotes {
color: #333333;
background-color: #FFFFFF;
border-bottom-style: none;

```

```
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
border-left-style: none;
border-left-width: 2px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 2px;
border-right-color: #D3D3D3;
}
```

```
#hitsdzdqpj .gt_sourcenote {
  font-size: 90%;
  padding-top: 4px;
  padding-bottom: 4px;
  padding-left: 5px;
  padding-right: 5px;
}
```

```
#hitsdzdqpj .gt_left {
  text-align: left;
}
```

```
#hitsdzdqpj .gt_center {
  text-align: center;
}
```

```
#hitsdzdqpj .gt_right {
  text-align: right;
  font-variant-numeric: tabular-nums;
}
```

```
#hitsdzdqpj .gt_font_normal {
  font-weight: normal;
}
```

```
#hitsdzdqpj .gt_font_bold {
  font-weight: bold;
}
```

```
#hitsdzdqpj .gt_font_italic {
  font-style: italic;
}
```

```
#hitsdzdqpj .gt_super {  
  font-size: 65%;  
}
```

```
#hitsdzdqpj .gt_footnote_marks {  
  font-size: 75%;  
  vertical-align: 0.4em;  
  position: initial;  
}
```

```
#hitsdzdqpj .gt_asterisk {  
  font-size: 100%;  
  vertical-align: 0;  
}
```

```
#hitsdzdqpj .gt_indent_1 {  
  text-indent: 5px;  
}
```

```
#hitsdzdqpj .gt_indent_2 {  
  text-indent: 10px;  
}
```

```
#hitsdzdqpj .gt_indent_3 {  
  text-indent: 15px;  
}
```

```
#hitsdzdqpj .gt_indent_4 {  
  text-indent: 20px;  
}
```

```
#hitsdzdqpj .gt_indent_5 {  
  text-indent: 25px;  
}
```

```
#hitsdzdqpj .katex-display {  
  display: inline-flex !important;  
  margin-bottom: 0.75em !important;  
}
```

```
#hitsdzdqpj div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-group:after {  
  height: 0px !important;  
}
```



```

</style>
<table class="gt_table" style="table-layout:fixed;width:100%;" data-quarto-disable-processing="false" data-
quarto-bootstrap="false">
  <colgroup>
    <col style="width:80%;"/>
    <col style="width:20%;"/>
  </colgroup>
  <thead>
    <tr class="gt_col_headings">
      <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col" id="labels">labels
      <th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col" id="varname">var
    </tr>
  </thead>
  <tbody class="gt_table_body">
    <tr><td headers="labels" class="gt_row gt_left">Skilled assistance during ANC</td>
<td headers="varname" class="gt_row gt_left">rh_anc_pvskill</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Attended 4+ ANC visits</td>
<td headers="varname" class="gt_row gt_left">rh_anc_4vs</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Number of months pregnant at time of first ANC visit</td>
<td headers="varname" class="gt_row gt_left">rh_anc_moprg</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Attended ANC &lt;4 months of pregnancy</td>
<td headers="varname" class="gt_row gt_left">rh_anc_4mo</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Took iron tablet/syrup during pregnancy of last birth</td>
<td headers="varname" class="gt_row gt_left">rh_anc_iron</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Took intestinal parasite drugs during pregnancy of last birth</td>
<td headers="varname" class="gt_row gt_left">rh_anc_parast</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Problem health care access: permission to go</td>
<td headers="varname" class="gt_row gt_left">rh_prob_permit</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Problem health care access: getting money</td>
<td headers="varname" class="gt_row gt_left">rh_prob_money</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Problem health care access: distance to facility</td>
<td headers="varname" class="gt_row gt_left">rh_prob_dist</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Problem health care access: not wanting to go alone</td>
<td headers="varname" class="gt_row gt_left">rh_prob_alone</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">At least one problem in accessing health care</td>
<td headers="varname" class="gt_row gt_left">rh_prob_minone</td></tr>
    <tr><td headers="labels" class="gt_row gt_left"></td>
<td headers="varname" class="gt_row gt_left">year</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Attended or completed at least secondary education</td>
<td headers="varname" class="gt_row gt_left">rc_edu_acceptable</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Literate - higher than secondary or can read part or whole sentence</td>
<td headers="varname" class="gt_row gt_left">rc_litr</td></tr>
    <tr><td headers="labels" class="gt_row gt_left">Accesses none of the three media at least once a week</td>

```

```

<td headers="varname" class="gt_row gt_left">rc_media_none</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Occupation among those employed in the past 12 months</td>
<td headers="varname" class="gt_row gt_left">rc_occup</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Work in agriculture in the past 12 months</td>
<td headers="varname" class="gt_row gt_left">rc_agri</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Have any health insurance</td>
<td headers="varname" class="gt_row gt_left">rc_hins_any</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Smokes any type of tobacco</td>
<td headers="varname" class="gt_row gt_left">rc_tobc_smk_any</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Birth weight less than 2.5 kg</td>
<td headers="varname" class="gt_row gt_left">ch_below_2p5</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">ARI symptoms in the 2 weeks before the survey</td>
<td headers="varname" class="gt_row gt_left">ch_ari</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Fever symptoms in the 2 weeks before the survey</td>
<td headers="varname" class="gt_row gt_left">ch_fever</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Started breastfeeding within one hour of birth - last-
born in the past 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_bf_start_1hr</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Started breastfeeding within one day of birth - last-
born in the past 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_bf_start_1day</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Received a prelacteal feed - last-born in the past 2 years ever breast fed</td>
<td headers="varname" class="gt_row gt_left">nt_bf_prelac_nobirthrecord</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Drank from a bottle with a nipple yesterday - under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_bottle</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Age-appropriately breastfed - last-born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_ageapp_bf</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given infant formula in day/night before survey - last-
born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_formula</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given other liquids in day/night before survey- last-
born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_liquids</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given fortified baby food in day/night before survey- last-
born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_bbyfood</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given grains in day/night before survey- last-
born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_grains</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given vitamin A rich food in day/night before survey- last-
born under 2 years</td>
<td headers="varname" class="gt_row gt_left">nt_vita</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child given other fruits or vegetables in day/night before survey- last-

```

born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_frtveg</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given roots or tubers in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_root</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given legumes or nuts in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_nuts</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given meat, fish, shellfish, or poultry in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_meatfish</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given eggs in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_eggs</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given cheese, yogurt, or other milk products in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_dairy</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given any solid or semisolid food in day/night before survey- last-born under 2 years</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_solids</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child given milk or milk products- last-born 6-23 months</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_fed\_milk</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child with minimum dietary diversity, 5 out of 8 food groups- last-born 6-23 months</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_mdd</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child with minimum meal frequency- last-born 6-23 months</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_mmf</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Child with minimum acceptable diet- last-born 6-23 months</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_mad</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Youngest children age 6-23 mos living with mother given Vit A rich food</td>  
<td headers="varname" class="gt\_row gt\_left">nt\_ch\_micro\_vaf</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">Current marital status</td>  
<td headers="varname" class="gt\_row gt\_left">ms\_mar\_stat</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">First marriage by age 15</td>  
<td headers="varname" class="gt\_row gt\_left">ms\_afm\_15</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">First marriage by age 18</td>  
<td headers="varname" class="gt\_row gt\_left">ms\_afm\_18</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">First marriage by age 20</td>  
<td headers="varname" class="gt\_row gt\_left">ms\_afm\_20</td></tr>  
<tr><td headers="labels" class="gt\_row gt\_left">First marriage by age 22</td>

```

<td headers="varname" class="gt_row gt_left">ms_afm_22</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First marriage by age 25</td>
<td headers="varname" class="gt_row gt_left">ms_afm_25</td></tr>
  <tr><td headers="labels" class="gt_row gt_left"></td>
<td headers="varname" class="gt_row gt_left">ms_age</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Never had sex</td>
<td headers="varname" class="gt_row gt_left">ms_sex_never</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First sex by age 15</td>
<td headers="varname" class="gt_row gt_left">ms_afs_15</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First sex by age 18</td>
<td headers="varname" class="gt_row gt_left">ms_afs_18</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First sex by age 20</td>
<td headers="varname" class="gt_row gt_left">ms_afs_20</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First sex by age 22</td>
<td headers="varname" class="gt_row gt_left">ms_afs_22</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">First sex by age 25</td>
<td headers="varname" class="gt_row gt_left">ms_afs_25</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Child's stool was disposed of appropriately among youngest children un
<td headers="varname" class="gt_row gt_left">ch_stool_safe</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Number of children under 12 years old</td>
<td headers="varname" class="gt_row gt_left">dm_children_under12</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">type of cooking fuel</td>
<td headers="varname" class="gt_row gt_left">dm_cooking_fuel</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife if she burns
<td headers="varname" class="gt_row gt_left">dm_dvjustify_burn</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife if she argues
<td headers="varname" class="gt_row gt_left">dm_dvjustify_argue</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife if she goes o
<td headers="varname" class="gt_row gt_left">dm_dvjustify_gooout</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife if she neglect
<td headers="varname" class="gt_row gt_left">dm_dvjustify_neglect</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife if she refuse
<td headers="varname" class="gt_row gt_left">dm_dvjustify_refusesex</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Agree that husband is justified in hitting or beating his wife for at least o
<td headers="varname" class="gt_row gt_left">dm_dvjustify_onereas</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">Believe a woman is justified to refuse sex with her husband if she knows
<td headers="varname" class="gt_row gt_left">dm_justify_refusesex</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">living in small town</td>
<td headers="varname" class="gt_row gt_left">dm_town</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">living in medium city</td>
<td headers="varname" class="gt_row gt_left">dm_city</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">living in capital city</td>
<td headers="varname" class="gt_row gt_left">dm_capital</td></tr>

```

```

<tr><td headers="labels" class="gt_row gt_left">poorest wealth quintile</td>
<td headers="varname" class="gt_row gt_left">dm_poorest</td></tr>
<tr><td headers="labels" class="gt_row gt_left">poorer wealth quintile</td>
<td headers="varname" class="gt_row gt_left">dm_poorer</td></tr>
<tr><td headers="labels" class="gt_row gt_left">middle wealth quintile</td>
<td headers="varname" class="gt_row gt_left">dm_middle</td></tr>
<tr><td headers="labels" class="gt_row gt_left">richer wealth quintile</td>
<td headers="varname" class="gt_row gt_left">dm_richer</td></tr>
<tr><td headers="labels" class="gt_row gt_left">richest wealth quintile</td>
<td headers="varname" class="gt_row gt_left">dm_richest</td></tr>
<tr><td headers="labels" class="gt_row gt_left">agricultural land (hectare)</td>
<td headers="varname" class="gt_row gt_left">dm_ag_land_ha</td></tr>
<tr><td headers="labels" class="gt_row gt_left">owns cattle</td>
<td headers="varname" class="gt_row gt_left">dm_cattle_own</td></tr>
<tr><td headers="labels" class="gt_row gt_left">owns goats</td>
<td headers="varname" class="gt_row gt_left">dm_goat_own</td></tr>
<tr><td headers="labels" class="gt_row gt_left">owns sheep</td>
<td headers="varname" class="gt_row gt_left">dm_sheep_own</td></tr>
<tr><td headers="labels" class="gt_row gt_left">owns poultry</td>
<td headers="varname" class="gt_row gt_left">dm_poultry_own</td></tr>
<tr><td headers="labels" class="gt_row gt_left">owns refrigerator</td>
<td headers="varname" class="gt_row gt_left">dm_refrigerator</td></tr>
<tr><td headers="labels" class="gt_row gt_left">female headed household</td>
<td headers="varname" class="gt_row gt_left">dm_female_headed</td></tr>
<tr><td headers="labels" class="gt_row gt_left">Age of head of household</td>
<td headers="varname" class="gt_row gt_left">dm_age_of_hh_head</td></tr>
<tr><td headers="labels" class="gt_row gt_left">Mild anemia - child 6-59 months</td>
<td headers="varname" class="gt_row gt_left">nt_ch_mild_anem</td></tr>
</tbody>

</table>
</div>

```

```

# ***** デバッグ用 *****
if (debug_mode) {
  cat("=== デバッグモード: 変数の詳細確認 ===\n")
  cat("共変量数:", length(covariate_names), "\n")
  cat("共変量:\n")

  # ダミーでない変数を抽出
  non_dummy <- list()
}

```

```

temp <- df_lasso[, covariate_names]
for (var in names(temp)) {
  if (!is_dummy_var(temp[[var]])) {
    non_dummy <- append(non_dummy, var)
  } else {
    cat("ダミー変数:", var)
    print(table(temp[[var]], useNA = "always"))
    cat("\n")
  }
}
cat("ダミー変数でない変数数:", length(non_dummy), "\n")
cat("ダミー変数でない変数:\n")
print(non_dummy)
}

```

=== デバッグモード: 変数の詳細確認 ===

共変量数: 84

共変量:

ダミー変数: rh\_anc\_pvskill

0 1 <NA>

14163 9951 0

ダミー変数: rh\_anc\_4vs

0 1 <NA>

817 23297 0

ダミー変数: rh\_anc\_4mo

0 1 <NA>

5689 18425 0

ダミー変数: rh\_anc\_iron

0 1 <NA>

2519 21595 0

ダミー変数: rh\_anc\_parast

0 1 <NA>

23482 632 0

ダミー変数: rh\_prob\_permit

0 1 <NA>

21098 3016 0

ダミー変数: rh\_prob\_money  
0 1 <NA>  
9638 14476 0

ダミー変数: rh\_prob\_dist  
0 1 <NA>  
14029 10085 0

ダミー変数: rh\_prob\_alone  
0 1 <NA>  
14905 9209 0

ダミー変数: rh\_prob\_minone  
0 1 <NA>  
5642 18472 0

ダミー変数: rc\_edu\_acceptable  
0 1 <NA>  
10833 13281 0

ダミー変数: rc\_litr  
0 1 <NA>  
1186 22928 0

ダミー変数: rc\_media\_none  
0 1 <NA>  
20481 3633 0

ダミー変数: rc\_agri  
0 1 <NA>  
20018 4096 0

ダミー変数: rc\_hins\_any  
0 1 <NA>  
5660 18454 0

ダミー変数: rc\_tobc\_smk\_any  
0 1 <NA>  
23677 437 0

ダミー変数: ch\_below\_2p5  
0 1 <NA>  
22637 1477 0

ダミー変数: ch\_ari

0 1 <NA>

22501 1613 0

ダミー変数: ch\_fever

0 1 <NA>

17596 6518 0

ダミー変数: nt\_bf\_start\_1hr

0 1 <NA>

10560 13554 0

ダミー変数: nt\_bf\_start\_1day

0 1 <NA>

1747 22367 0

ダミー変数: nt\_bf\_prelac\_nobirthrecord

0 1 <NA>

6127 17987 0

ダミー変数: nt\_bottle

0 1 <NA>

12186 11928 0

ダミー変数: nt\_ageapp\_bf

0 1 <NA>

113 24001 0

ダミー変数: nt\_formula

0 1 <NA>

21147 2967 0

ダミー変数: nt\_liquids

0 1 <NA>

6251 17863 0

ダミー変数: nt\_bbyfood

0 <NA>

24114 0

ダミー変数: nt\_grains

0 1 <NA>



2620 21494 0

ダミー変数: nt\_vita

0 1 <NA>

5530 18584 0

ダミー変数: nt\_frtveg

0 1 <NA>

7177 16937 0

ダミー変数: nt\_root

0 1 <NA>

4822 19292 0

ダミー変数: nt\_nuts

0 1 <NA>

13504 10610 0

ダミー変数: nt\_meatfish

0 1 <NA>

4144 19970 0

ダミー変数: nt\_eggs

0 1 <NA>

10086 14028 0

ダミー変数: nt\_dairy

0 1 <NA>

14105 10009 0

ダミー変数: nt\_solids

1 <NA>

24114 0

ダミー変数: nt\_fed\_milk

0 1 <NA>

3488 20626 0

ダミー変数: nt\_mdd

0 1 <NA>

5285 18829 0

ダミー変数: nt\_mmf

0 1 <NA>  
1709 22405 0

ダミー変数: nt\_mad  
0 1 <NA>  
7861 16253 0

ダミー変数: nt\_ch\_micro\_vaf  
1 <NA>  
24114 0

ダミー変数: ms\_afm\_15  
0 1 <NA>  
22967 1147 0

ダミー変数: ms\_afm\_18  
0 1 <NA>  
16921 7193 0

ダミー変数: ms\_afm\_20  
0 1 <NA>  
11971 12143 0

ダミー変数: ms\_afm\_22  
0 1 <NA>  
8297 15817 0

ダミー変数: ms\_afm\_25  
0 1 <NA>  
4983 19131 0

ダミー変数: ms\_sex\_never  
0 <NA>  
24114 0

ダミー変数: ms\_afs\_15  
0 1 <NA>  
21295 2819 0

ダミー変数: ms\_afs\_18  
0 1 <NA>  
10943 13171 0

ダミー変数: ms\_afs\_20

0 1 <NA>

5333 18781 0

ダミー変数: ms\_afs\_22

0 1 <NA>

2655 21459 0

ダミー変数: ms\_afs\_25

0 1 <NA>

1047 23067 0

ダミー変数: ch\_stool\_safe

0 1 <NA>

19002 5112 0

ダミー変数: dm\_dvjustify\_burn

0 1 <NA>

23901 213 0

ダミー変数: dm\_dvjustify\_argue

0 1 <NA>

23931 183 0

ダミー変数: dm\_dvjustify\_goout

0 1 <NA>

23832 282 0

ダミー変数: dm\_dvjustify\_neglect

0 1 <NA>

23549 565 0

ダミー変数: dm\_dvjustify\_refusesex

0 1 <NA>

23965 149 0

ダミー変数: dm\_dvjustify\_onereas

0 1 <NA>

23257 857 0

ダミー変数: dm\_justify\_refusesex

0 1 <NA>

985 23129 0

ダミー変数: dm\_town

0 1 <NA>

18197 5917 0

ダミー変数: dm\_city

0 1 <NA>

16137 7977 0

ダミー変数: dm\_capital

0 1 <NA>

21705 2409 0

ダミー変数: dm\_poorest

0 1 <NA>

18934 5180 0

ダミー変数: dm\_poorer

0 1 <NA>

17189 6925 0

ダミー変数: dm\_middle

0 1 <NA>

18364 5750 0

ダミー変数: dm\_richer

0 1 <NA>

20237 3877 0

ダミー変数: dm\_richest

0 1 <NA>

21732 2382 0

ダミー変数: dm\_cattle\_own

0 1 <NA>

21280 2834 0

ダミー変数: dm\_goat\_own

0 1 <NA>

23603 511 0

ダミー変数: dm\_sheep\_own

0 1 <NA>

21701 2413 0

ダミー変数: dm\_poultry\_own

0 1 <NA>

14395 9719 0

ダミー変数: dm\_refrigerator

0 1 <NA>

14429 9685 0

ダミー変数: dm\_female\_headed

0 1 <NA>

19954 4160 0

ダミー変数: nt\_ch\_mild\_anem

0 1 <NA>

16585 7529 0

ダミー変数でない変数数: 9

ダミー変数でない変数:

[[1]]

[1] "rh\_anc\_moprg"

[[2]]

[1] "year"

[[3]]

[1] "rc\_occup"

[[4]]

[1] "ms\_mar\_stat"

[[5]]

[1] "ms\_age"

[[6]]

[1] "dm\_children\_under12"

[[7]]

[1] "dm\_cooking\_fuel"

[[8]]

[1] "dm\_ag\_land\_ha"

```
[[9]]  
[1] "dm_age_of_hh_head"
```

```
# ***** デバッグ用 *****
```

### 3. LASSOによる変数選択（介入前データ）

```
library(gt)
```

```
# =====  
# 3. LASSO変数選択（介入前データ）  
# =====  
# 完全に一から再開  
cat("=== データ処理を再開 ===\n")
```

```
=== データ処理を再開 ===
```

```
cat("df_lasso の次元:", dim(df_lasso), "\n")
```

```
df_lasso の次元: 24114 182
```

```
cat("covariate_names の数:", length(covariate_names), "\n")
```

```
covariate_names の数: 84
```

```
# Step 1: model.matrix作成
```

```
cat("Step 1: model.matrix作成中...\n")
```

```
Step 1: model.matrix作成中...
```

```
X_matrix <- model.matrix(~ . - 1, data = df_lasso[, covariate_names, drop=FALSE])
```

```
Y <- df_lasso$nt_ch_stunt
```

```
cat("=== 問題列の除去 ===\n")
```

```
=== 問題列の除去 ===
```

```
# Step 1: 標準偏差が0の列を特定・除去
col_sds <- apply(X_matrix, 2, sd, na.rm = TRUE)
zero_sd_cols <- is.na(col_sds) | col_sds == 0
cat("除去する定数列の数:", sum(zero_sd_cols), "\n")
```

除去する定数列の数: 4

```
if(sum(zero_sd_cols) > 0) {
  cat("除去する列名:\n")
  print(colnames(X_matrix)[zero_sd_cols])

  # 定数列を除去
  X_matrix_clean <- X_matrix[, !zero_sd_cols, drop = FALSE]
} else {
  X_matrix_clean <- X_matrix
}
```

除去する列名:

```
[1] "nt_bbyfood"    "nt_solids"     "nt_ch_micro_vaf" "ms_sex_never"
```

```
cat("列除去後のX_matrix次元:", dim(X_matrix_clean), "\n")
```

列除去後のX\_matrix次元: 24114 89

```
# Step 2: スケーリング実行
cat("=== クリーンなデータでスケーリング ===\n")
```

=== クリーンなデータでスケーリング ===

```
X_scaled <- scale(X_matrix_clean)
cat("スケーリング後の次元:", dim(X_scaled), "\n")
```

スケーリング後の次元: 24114 89

```
cat("スケーリング後のNA数:", sum(is.na(X_scaled)), "\n")
```

スケーリング後のNA数: 0

```

# Step 3: まだNAがある場合の詳細チェック
if(sum(is.na(X_scaled)) > 0) {
  cat("=== 残存するNA問題の詳細分析 ===\n")

  # 各列のNA数をチェック
  col_na_counts <- apply(X_scaled, 2, function(x) sum(is.na(x)))
  problematic_cols <- col_na_counts > 0

  cat("NA値を含む列数:", sum(problematic_cols), "\n")
  if(sum(problematic_cols) > 0) {
    cat("問題のある列:\n")
    print(col_na_counts[problematic_cols])

    # 問題のある列を除去
    X_scaled <- X_scaled[, !problematic_cols, drop = FALSE]
    cat("問題列除去後の次元:", dim(X_scaled), "\n")
  }
}

# Step 4: 最終確認
cat("=== 最終確認 ===\n")

```

=== 最終確認 ===

```
cat("X_scaled次元:", dim(X_scaled), "\n")
```

X\_scaled次元: 24114 89

```
cat("X_scaledのNA数:", sum(is.na(X_scaled)), "\n")
```

X\_scaledのNA数: 0

```
cat("Y長さ:", length(Y), "\n")
```

Y長さ: 24114

```
cat("Y分布:", table(Y), "\n")
```

Y分布: 19824 4290



```
# データが正常な場合のLASSO実行
if(nrow(X_scaled) > 0 && ncol(X_scaled) > 0 && sum(is.na(X_scaled)) == 0) {
  cat("=== LASSO実行 ===\n")
  set.seed(123)

  cv_model <- cv.glmnet(X_scaled, Y,
    alpha = 1,
    family = "multinomial",
    nfolds = 3)

  cat("LASSO実行成功！\n")
} else {
  cat("まだデータに問題があります\n")
}
```

=== LASSO実行 ===  
LASSO実行成功！

```
cat("最適 λ (lambda.min):", cv_model$lambda.min, "\n")
```

最適 λ (lambda.min): 0.001559054

```
cat("1se λ (lambda.1se):", cv_model$lambda.1se, "\n\n")
```

1se λ (lambda.1se): 0.004761126

```
# 選択された変数の抽出
coef_min <- coef(cv_model, s = "lambda.min")
selected_vars <- c()
for(i in 1:length(coef_min)) {
  coef_matrix <- as.matrix(coef_min[[i]])
  class_vars <- rownames(coef_matrix)[coef_matrix[,1] != 0 & rownames(coef_matrix) != "(Intercept)"]
  selected_vars <- union(selected_vars, class_vars)
}

cat("LASSO選択変数数:", length(selected_vars), "\n")
```

LASSO選択変数数: 62

```

if(length(selected_vars) > 0) {
  cat("選択された変数:\n")
  for(i in 1:length(selected_vars)) {
    cat(i, ":", selected_vars[i], "\n")
  }
  # 変数の重要度確認
  coef_importance <- abs(as.matrix(coef(cv_model, s = "lambda.min"))[[1]])
  top_vars <- head(sort(coef_importance[, 1], decreasing = TRUE), 20)
  coef_importance <- coef_importance %>%
    as_tibble(rownames = "labels") %>%
    filter(labels %in% selected_vars) %>%
    left_join(var_labels, by = join_by("labels" == "varname"))
  saveRDS(coef_importance, file.path(save_path, "lasso_selected_vars_importance.rds"))

  # 列幅を指定して結果出力
  coef_importance %>%
    gt() %>%
    cols_width(
      1 ~ pct(15),
      2 ~ pct(15),
      3 ~ pct(70)
    ) %>%
    fmt_number(
      columns = c("lambda.min"),
      decimals = 4 # 小数点以下4桁
    ) %>%
    print()
} else {
  stop("変数が選択されませんでした。λの値を調整してください。")
}

```

選択された変数:

```

1 : rh_anc_pvskill
2 : rh_anc_4vs
3 : rh_anc_4mo
4 : rh_anc_iron
5 : rh_anc_parast
6 : rh_prob_permit
7 : rh_prob_dist
8 : rh_prob_minone
9 : year2007
10 : year2008

```

11 : year2009  
12 : year2010  
13 : year2011  
14 : year2014  
15 : year2015  
16 : year2016  
17 : rc\_edu\_acceptable  
18 : rc\_litr  
19 : rc\_media\_none  
20 : rc\_occup  
21 : rc\_agri  
22 : rc\_hins\_any  
23 : rc\_tobc\_smk\_any  
24 : ch\_below\_2p5  
25 : nt\_bf\_start\_1hr  
26 : nt\_bf\_start\_1day  
27 : nt\_bf\_prelac\_nobirthrecord  
28 : nt\_bottle  
29 : nt\_formula  
30 : nt\_liquids  
31 : nt\_grains  
32 : nt\_frtveg  
33 : nt\_root  
34 : nt\_nuts  
35 : nt\_eggs  
36 : nt\_dairy  
37 : nt\_fed\_milk  
38 : nt\_mdd  
39 : nt\_mmf  
40 : ms\_mar\_stat  
41 : ms\_afm\_15  
42 : ms\_afs\_15  
43 : ms\_afs\_18  
44 : ms\_afs\_25  
45 : ch\_stool\_safe  
46 : dm\_children\_under12  
47 : dm\_cooking\_fuel  
48 : dm\_dvjustify\_argue  
49 : dm\_dvjustify\_neglect  
50 : dm\_justify\_refusesex  
51 : dm\_town  
52 : dm\_capital  
53 : dm\_poorest

```

54 : dm_poorer
55 : dm_richer
56 : dm_richest
57 : dm_ag_land_ha
58 : dm_cattle_own
59 : dm_goat_own
60 : dm_sheep_own
61 : dm_refrigerator
62 : dm_age_of_hh_head
<div id="qrwwcuyfmt" style="padding-left:0px;padding-right:0px;padding-top:10px;padding-bottom:10px;overflow-
x:auto;overflow-y:auto;width:auto;height:auto;">
  <style>#qrwwcuyfmt table {
    font-family: system-ui, 'Segoe UI', Roboto, Helvetica, Arial, sans-serif, 'Apple Color Emoji', 'Segoe UI Emoji', 'Segoe UI Symbol';
    -webkit-font-smoothing: antialiased;
    -moz-osx-font-smoothing: grayscale;
  }

  #qrwwcuyfmt thead, #qrwwcuyfmt tbody, #qrwwcuyfmt tfoot, #qrwwcuyfmt tr, #qrwwcuyfmt td, #qrwwcuyfmt th {
    border-style: none;
  }

  #qrwwcuyfmt p {
    margin: 0;
    padding: 0;
  }

  #qrwwcuyfmt .gt_table {
    display: table;
    border-collapse: collapse;
    line-height: normal;
    margin-left: auto;
    margin-right: auto;
    color: #333333;
    font-size: 16px;
    font-weight: normal;
    font-style: normal;
    background-color: #FFFFFF;
    width: auto;
    border-top-style: solid;
    border-top-width: 2px;
    border-top-color: #A8A8A8;
    border-right-style: none;
    border-right-width: 2px;

```

```
border-right-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #A8A8A8;
border-left-style: none;
border-left-width: 2px;
border-left-color: #D3D3D3;
}
```

```
#qrwwcuyfmt .gt_caption {
padding-top: 4px;
padding-bottom: 4px;
}
```

```
#qrwwcuyfmt .gt_title {
color: #333333;
font-size: 125%;
font-weight: initial;
padding-top: 4px;
padding-bottom: 4px;
padding-left: 5px;
padding-right: 5px;
border-bottom-color: #FFFFFF;
border-bottom-width: 0;
}
```

```
#qrwwcuyfmt .gt_subtitle {
color: #333333;
font-size: 85%;
font-weight: initial;
padding-top: 3px;
padding-bottom: 5px;
padding-left: 5px;
padding-right: 5px;
border-top-color: #FFFFFF;
border-top-width: 0;
}
```

```
#qrwwcuyfmt .gt_heading {
background-color: #FFFFFF;
text-align: center;
border-bottom-color: #FFFFFF;
border-left-style: none;
```

```

border-left-width: 1px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 1px;
border-right-color: #D3D3D3;
}

#qrwwcuyfmt .gt_bottom_border {
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
}

#qrwwcuyfmt .gt_col_headings {
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
border-left-style: none;
border-left-width: 1px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 1px;
border-right-color: #D3D3D3;
}

#qrwwcuyfmt .gt_col_heading {
color: #333333;
background-color: #FFFFFF;
font-size: 100%;
font-weight: normal;
text-transform: inherit;
border-left-style: none;
border-left-width: 1px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 1px;
border-right-color: #D3D3D3;
vertical-align: bottom;
padding-top: 5px;
padding-bottom: 6px;
}

```

```

padding-left: 5px;
padding-right: 5px;
overflow-x: hidden;
}

#qrwwcuyfmt.gt_column_spanner_outer {
color: #333333;
background-color: #FFFFFF;
font-size: 100%;
font-weight: normal;
text-transform: inherit;
padding-top: 0;
padding-bottom: 0;
padding-left: 4px;
padding-right: 4px;
}

#qrwwcuyfmt.gt_column_spanner_outer:first-child {
padding-left: 0;
}

#qrwwcuyfmt.gt_column_spanner_outer:last-child {
padding-right: 0;
}

#qrwwcuyfmt.gt_column_spanner {
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
vertical-align: bottom;
padding-top: 5px;
padding-bottom: 5px;
overflow-x: hidden;
display: inline-block;
width: 100%;
}

#qrwwcuyfmt.gt_spanner_row {
border-bottom-style: hidden;
}

#qrwwcuyfmt.gt_group_heading {
padding-top: 8px;

```

```

padding-bottom: 8px;
padding-left: 5px;
padding-right: 5px;
color: #333333;
background-color: #FFFFFF;
font-size: 100%;
font-weight: initial;
text-transform: inherit;
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
border-left-style: none;
border-left-width: 1px;
border-left-color: #D3D3D3;
border-right-style: none;
border-right-width: 1px;
border-right-color: #D3D3D3;
vertical-align: middle;
text-align: left;
}

#qrwwcuyfmt .gt_empty_group_heading {
padding: 0.5px;
color: #333333;
background-color: #FFFFFF;
font-size: 100%;
font-weight: initial;
border-top-style: solid;
border-top-width: 2px;
border-top-color: #D3D3D3;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
vertical-align: middle;
}

#qrwwcuyfmt .gt_from_md > :first-child {
margin-top: 0;
}

```



```
#qrwwcuyfmt .gt_from_md > :last-child {  
  margin-bottom: 0;  
}
```

```
#qrwwcuyfmt .gt_row {  
  padding-top: 8px;  
  padding-bottom: 8px;  
  padding-left: 5px;  
  padding-right: 5px;  
  margin: 10px;  
  border-top-style: solid;  
  border-top-width: 1px;  
  border-top-color: #D3D3D3;  
  border-left-style: none;  
  border-left-width: 1px;  
  border-left-color: #D3D3D3;  
  border-right-style: none;  
  border-right-width: 1px;  
  border-right-color: #D3D3D3;  
  vertical-align: middle;  
  overflow-x: hidden;  
}
```

```
#qrwwcuyfmt .gt_stub {  
  color: #333333;  
  background-color: #FFFFFF;  
  font-size: 100%;  
  font-weight: initial;  
  text-transform: inherit;  
  border-right-style: solid;  
  border-right-width: 2px;  
  border-right-color: #D3D3D3;  
  padding-left: 5px;  
  padding-right: 5px;  
}
```

```
#qrwwcuyfmt .gt_stub_row_group {  
  color: #333333;  
  background-color: #FFFFFF;  
  font-size: 100%;  
  font-weight: initial;  
  text-transform: inherit;  
  border-right-style: solid;
```

```

border-right-width: 2px;
border-right-color: #D3D3D3;
padding-left: 5px;
padding-right: 5px;
vertical-align: top;
}

#qrwwcuyfmt .gt_row_group_first td {
border-top-width: 2px;
}

#qrwwcuyfmt .gt_row_group_first th {
border-top-width: 2px;
}

#qrwwcuyfmt .gt_summary_row {
color: #333333;
background-color: #FFFFFF;
text-transform: inherit;
padding-top: 8px;
padding-bottom: 8px;
padding-left: 5px;
padding-right: 5px;
}

#qrwwcuyfmt .gt_first_summary_row {
border-top-style: solid;
border-top-color: #D3D3D3;
}

#qrwwcuyfmt .gt_first_summary_row.thick {
border-top-width: 2px;
}

#qrwwcuyfmt .gt_last_summary_row {
padding-top: 8px;
padding-bottom: 8px;
padding-left: 5px;
padding-right: 5px;
border-bottom-style: solid;
border-bottom-width: 2px;
border-bottom-color: #D3D3D3;
}

```

```

#qrwwcuyfmt .gt_grand_summary_row {
  color: #333333;
  background-color: #FFFFFF;
  text-transform: inherit;
  padding-top: 8px;
  padding-bottom: 8px;
  padding-left: 5px;
  padding-right: 5px;
}

#qrwwcuyfmt .gt_first_grand_summary_row {
  padding-top: 8px;
  padding-bottom: 8px;
  padding-left: 5px;
  padding-right: 5px;
  border-top-style: double;
  border-top-width: 6px;
  border-top-color: #D3D3D3;
}

#qrwwcuyfmt .gt_last_grand_summary_row_top {
  padding-top: 8px;
  padding-bottom: 8px;
  padding-left: 5px;
  padding-right: 5px;
  border-bottom-style: double;
  border-bottom-width: 6px;
  border-bottom-color: #D3D3D3;
}

#qrwwcuyfmt .gt_stripped {
  background-color: rgba(128, 128, 128, 0.05);
}

#qrwwcuyfmt .gt_table_body {
  border-top-style: solid;
  border-top-width: 2px;
  border-top-color: #D3D3D3;
  border-bottom-style: solid;
  border-bottom-width: 2px;
  border-bottom-color: #D3D3D3;
}

```

```
#qrwwcuyfmt .gt_footnotes {  
  color: #333333;  
  background-color: #FFFFFF;  
  border-bottom-style: none;  
  border-bottom-width: 2px;  
  border-bottom-color: #D3D3D3;  
  border-left-style: none;  
  border-left-width: 2px;  
  border-left-color: #D3D3D3;  
  border-right-style: none;  
  border-right-width: 2px;  
  border-right-color: #D3D3D3;  
}
```

```
#qrwwcuyfmt .gt_footnote {  
  margin: 0px;  
  font-size: 90%;  
  padding-top: 4px;  
  padding-bottom: 4px;  
  padding-left: 5px;  
  padding-right: 5px;  
}
```

```
#qrwwcuyfmt .gt_sourcenotes {  
  color: #333333;  
  background-color: #FFFFFF;  
  border-bottom-style: none;  
  border-bottom-width: 2px;  
  border-bottom-color: #D3D3D3;  
  border-left-style: none;  
  border-left-width: 2px;  
  border-left-color: #D3D3D3;  
  border-right-style: none;  
  border-right-width: 2px;  
  border-right-color: #D3D3D3;  
}
```

```
#qrwwcuyfmt .gt_sourcenote {  
  font-size: 90%;  
  padding-top: 4px;  
  padding-bottom: 4px;  
  padding-left: 5px;
```

```

padding-right: 5px;
}

#qrwwcuyfmt .gt_left {
text-align: left;
}

#qrwwcuyfmt .gt_center {
text-align: center;
}

#qrwwcuyfmt .gt_right {
text-align: right;
font-variant-numeric: tabular-nums;
}

#qrwwcuyfmt .gt_font_normal {
font-weight: normal;
}

#qrwwcuyfmt .gt_font_bold {
font-weight: bold;
}

#qrwwcuyfmt .gt_font_italic {
font-style: italic;
}

#qrwwcuyfmt .gt_super {
font-size: 65%;
}

#qrwwcuyfmt .gt_footnote_marks {
font-size: 75%;
vertical-align: 0.4em;
position: initial;
}

#qrwwcuyfmt .gt_asterisk {
font-size: 100%;
vertical-align: 0;
}

```

```
#qrwwcuyfmt .gt_indent_1 {
  text-indent: 5px;
}
```

```
#qrwwcuyfmt .gt_indent_2 {
  text-indent: 10px;
}
```

```
#qrwwcuyfmt .gt_indent_3 {
  text-indent: 15px;
}
```

```
#qrwwcuyfmt .gt_indent_4 {
  text-indent: 20px;
}
```

```
#qrwwcuyfmt .gt_indent_5 {
  text-indent: 25px;
}
```

```
#qrwwcuyfmt .katex-display {
  display: inline-flex !important;
  margin-bottom: 0.75em !important;
}
```

```
#qrwwcuyfmt div.Reactable > div.rt-table > div.rt-thead > div.rt-tr.rt-tr-group-header > div.rt-th-
group:after {
  height: 0px !important;
}
```

```
</style>
```

```
<table class="gt_table" style="table-layout:fixed;width:100%;" data-quarto-disable-processing="false" data-
quarto-bootstrap="false">
```

```
<colgroup>
```

```
<col style="width:15%;"/>
```

```
<col style="width:15%;"/>
```

```
<col style="width:70%;"/>
```

```
</colgroup>
```

```
<thead>
```

```
<tr class="gt_col_headings">
```

```
<th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col" id="labels">labels
```

```
<th class="gt_col_heading gt_columns_bottom_border gt_right" rowspan="1" colspan="1" scope="col" id="lambda.min">
```

```
<th class="gt_col_heading gt_columns_bottom_border gt_left" rowspan="1" colspan="1" scope="col" id="labels.y">labe
```

```
</tr>
```

```

</thead>
<tbody class="gt_table_body">
  <tr><td headers="labels" class="gt_row gt_left">rh_anc_pvskill</td>
<td headers="lambda.min" class="gt_row gt_right">0.0148</td>
<td headers="labels.y" class="gt_row gt_left">Skilled assistance during ANC</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_anc_4vs</td>
<td headers="lambda.min" class="gt_row gt_right">0.0065</td>
<td headers="labels.y" class="gt_row gt_left">Attended 4+ ANC visits</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_anc_4mo</td>
<td headers="lambda.min" class="gt_row gt_right">0.0166</td>
<td headers="labels.y" class="gt_row gt_left">Attended ANC &lt;4 months of pregnancy</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_anc_iron</td>
<td headers="lambda.min" class="gt_row gt_right">0.0003</td>
<td headers="labels.y" class="gt_row gt_left">Took iron tablet/syrup during pregnancy of last birth</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_anc_parast</td>
<td headers="lambda.min" class="gt_row gt_right">0.0076</td>
<td headers="labels.y" class="gt_row gt_left">Took intestinal parasite drugs during pregnancy of last birth</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_prob_permit</td>
<td headers="lambda.min" class="gt_row gt_right">0.0142</td>
<td headers="labels.y" class="gt_row gt_left">Problem health care access: permission to go</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_prob_dist</td>
<td headers="lambda.min" class="gt_row gt_right">0.0058</td>
<td headers="labels.y" class="gt_row gt_left">Problem health care access: distance to facility</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rh_prob_minone</td>
<td headers="lambda.min" class="gt_row gt_right">0.0128</td>
<td headers="labels.y" class="gt_row gt_left">At least one problem in accessing health care</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2007</td>
<td headers="lambda.min" class="gt_row gt_right">0.0295</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2008</td>
<td headers="lambda.min" class="gt_row gt_right">0.0323</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2009</td>
<td headers="lambda.min" class="gt_row gt_right">0.0245</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2010</td>
<td headers="lambda.min" class="gt_row gt_right">0.0165</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2011</td>
<td headers="lambda.min" class="gt_row gt_right">0.0251</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2014</td>
<td headers="lambda.min" class="gt_row gt_right">0.0144</td>

```

```

<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2015</td>
<td headers="lambda.min" class="gt_row gt_right">0.0213</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">year2016</td>
<td headers="lambda.min" class="gt_row gt_right">0.0410</td>
<td headers="labels.y" class="gt_row gt_left">NA</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_edu_acceptable</td>
<td headers="lambda.min" class="gt_row gt_right">0.0755</td>
<td headers="labels.y" class="gt_row gt_left">Attended or completed at least secondary education</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_litr</td>
<td headers="lambda.min" class="gt_row gt_right">0.0415</td>
<td headers="labels.y" class="gt_row gt_left">Literate - higher than secondary or can read part or whole sentence</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_media_none</td>
<td headers="lambda.min" class="gt_row gt_right">0.0115</td>
<td headers="labels.y" class="gt_row gt_left">Accesses none of the three media at least once a week</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_occup</td>
<td headers="lambda.min" class="gt_row gt_right">0.0067</td>
<td headers="labels.y" class="gt_row gt_left">Occupation among those employed in the past 12 months</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_agri</td>
<td headers="lambda.min" class="gt_row gt_right">0.0269</td>
<td headers="labels.y" class="gt_row gt_left">Work in agriculture in the past 12 months</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_hins_any</td>
<td headers="lambda.min" class="gt_row gt_right">0.0104</td>
<td headers="labels.y" class="gt_row gt_left">Have any health insurance</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">rc_tobc_smk_any</td>
<td headers="lambda.min" class="gt_row gt_right">0.0139</td>
<td headers="labels.y" class="gt_row gt_left">Smokes any type of tobacco</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">ch_below_2p5</td>
<td headers="lambda.min" class="gt_row gt_right">0.1341</td>
<td headers="labels.y" class="gt_row gt_left">Birth weight less than 2.5 kg</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">nt_bf_start_1hr</td>
<td headers="lambda.min" class="gt_row gt_right">0.0225</td>
<td headers="labels.y" class="gt_row gt_left">Started breastfeeding within one hour of birth - last-
born in the past 2 years</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">nt_bf_start_1day</td>
<td headers="lambda.min" class="gt_row gt_right">0.0320</td>
<td headers="labels.y" class="gt_row gt_left">Started breastfeeding within one day of birth - last-
born in the past 2 years</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">nt_bf_prelac_nobirthrecord</td>
<td headers="lambda.min" class="gt_row gt_right">0.0026</td>
<td headers="labels.y" class="gt_row gt_left">Received a prelacteal feed - last-born in the past 2 years ever breast fed (with
  <tr><td headers="labels" class="gt_row gt_left">nt_bottle</td>

```



|            |                                                                                                  |
|------------|--------------------------------------------------------------------------------------------------|
| lambda.min | 0.0261                                                                                           |
| labels.y   | Drank from a bottle with a nipple yesterday - under 2 years                                      |
| labels     | nt_formula                                                                                       |
| lambda.min | 0.0011                                                                                           |
| labels.y   | Child given infant formula in day/night before survey - last-born under 2 years                  |
| labels     | nt_liquids                                                                                       |
| lambda.min | 0.0050                                                                                           |
| labels.y   | Child given other liquids in day/night before survey- last-born under 2 years                    |
| labels     | nt_grains                                                                                        |
| lambda.min | 0.0132                                                                                           |
| labels.y   | Child given grains in day/night before survey- last-born under 2 years                           |
| labels     | nt_frtveg                                                                                        |
| lambda.min | 0.0095                                                                                           |
| labels.y   | Child given other fruits or vegetables in day/night before survey- last-born under 2 years       |
| labels     | nt_root                                                                                          |
| lambda.min | 0.0086                                                                                           |
| labels.y   | Child given roots or tubers in day/night before survey- last-born under 2 years                  |
| labels     | nt_nuts                                                                                          |
| lambda.min | 0.0054                                                                                           |
| labels.y   | Child given legumes or nuts in day/night before survey- last-born under 2 years                  |
| labels     | nt_eggs                                                                                          |
| lambda.min | 0.0076                                                                                           |
| labels.y   | Child given eggs in day/night before survey- last-born under 2 years                             |
| labels     | nt_dairy                                                                                         |
| lambda.min | 0.0200                                                                                           |
| labels.y   | Child given cheese, yogurt, or other milk products in day/night before survey born under 2 years |
| labels     | nt_fed_milk                                                                                      |
| lambda.min | 0.0570                                                                                           |
| labels.y   | Child given milk or milk products- last-born 6-23 months                                         |
| labels     | nt_mdd                                                                                           |
| lambda.min | 0.0010                                                                                           |
| labels.y   | Child with minimum dietary diversity, 5 out of 8 food groups- last-born 6-23 months              |
| labels     | nt_mmf                                                                                           |
| lambda.min | 0.0062                                                                                           |

|                                                                                              |        |
|----------------------------------------------------------------------------------------------|--------|
| Child with minimum meal frequency- last-born 6-23 months                                     |        |
| ms_mar_stat                                                                                  | 0.0063 |
| Current marital status                                                                       |        |
| ms_afm_15                                                                                    | 0.0093 |
| First marriage by age 15                                                                     |        |
| ms_afs_15                                                                                    | 0.0029 |
| First sex by age 15                                                                          |        |
| ms_afs_18                                                                                    | 0.0076 |
| First sex by age 18                                                                          |        |
| ms_afs_25                                                                                    | 0.0014 |
| First sex by age 25                                                                          |        |
| ch_stool_safe                                                                                | 0.0061 |
| Child's stool was disposed of appropriately among youngest children under age 12             |        |
| dm_children_under12                                                                          | 0.0805 |
| Number of children under 12 years old                                                        |        |
| dm_cooking_fuel                                                                              | 0.0097 |
| type of cooking fuel                                                                         |        |
| dm_dvjustify_argue                                                                           | 0.0007 |
| Agree that husband is justified in hitting or beating his wife if she argues with him        |        |
| dm_dvjustify_neglect                                                                         | 0.0001 |
| Agree that husband is justified in hitting or beating his wife if she neglects the household |        |
| dm_justify_refusesex                                                                         | 0.0091 |
| Believe a woman is justified to refuse sex with her husband if she knows he's violent        |        |
| dm_town                                                                                      | 0.0111 |
| living in small town                                                                         |        |
| dm_capital                                                                                   | 0.0364 |
| living in capital city                                                                       |        |
| dm_poorrest                                                                                  | 0.0951 |

```

<td headers="labels.y" class="gt_row gt_left">poorest wealth quintile</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_poorer</td>
<td headers="lambda.min" class="gt_row gt_right">0.0277</td>
<td headers="labels.y" class="gt_row gt_left">poorer wealth quintile</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_richer</td>
<td headers="lambda.min" class="gt_row gt_right">0.0536</td>
<td headers="labels.y" class="gt_row gt_left">richer wealth quintile</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_richest</td>
<td headers="lambda.min" class="gt_row gt_right">0.0904</td>
<td headers="labels.y" class="gt_row gt_left">richest wealth quintile</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_ag_land_ha</td>
<td headers="lambda.min" class="gt_row gt_right">0.0040</td>
<td headers="labels.y" class="gt_row gt_left">agricultural land (hectare)</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_cattle_own</td>
<td headers="lambda.min" class="gt_row gt_right">0.0170</td>
<td headers="labels.y" class="gt_row gt_left">owns cattle</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_goat_own</td>
<td headers="lambda.min" class="gt_row gt_right">0.0025</td>
<td headers="labels.y" class="gt_row gt_left">owns goats</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_sheep_own</td>
<td headers="lambda.min" class="gt_row gt_right">0.0500</td>
<td headers="labels.y" class="gt_row gt_left">owns sheep</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_refrigerator</td>
<td headers="lambda.min" class="gt_row gt_right">0.0595</td>
<td headers="labels.y" class="gt_row gt_left">owns refrigerator</td></tr>
  <tr><td headers="labels" class="gt_row gt_left">dm_age_of_hh_head</td>
<td headers="lambda.min" class="gt_row gt_right">0.0237</td>
<td headers="labels.y" class="gt_row gt_left">Age of head of household</td></tr>
</tbody>

</table>
</div>

```

4. 多項GLM-PSM（介入前データによる学習）
5. PSM結果のバランス診断
6. PSMのバランスチェック（クラスターサンプリングによる重み付けなし）
7. DID分析結果のまとめ
9. 結果の保存・可視化